

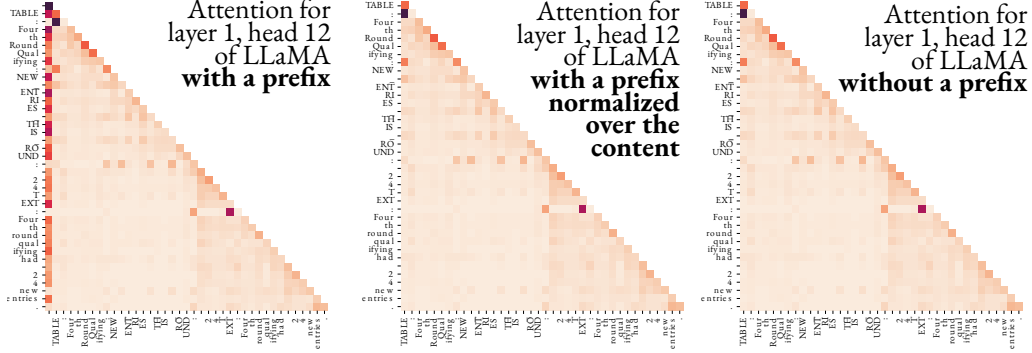


Figure 5: The attention of the twelfth head of the first layer of LLaMA (Touvron et al., 2023). The left plot shows the attention with a prefix of length one. The second plot shows the same attention but normalized such that the attention over the non-prefix positions sums to 1. The right plot shows the attention of the pre-trained model (without prefix). The center and the right plots are the same, illustrating that the presence of the prefix indeed only scales down the attention over the content (non-prefix positions) but does not change its relative distribution, providing empirical validation of Equation (6). The test sequence is TABLE: Fourth Round Qualifying : NEW_ENTRIES_THIS_ROUND : 24 TEXT: Fourth round qualifying had 24 new entries. from the DART table-to-test dataset (Nan et al., 2021).

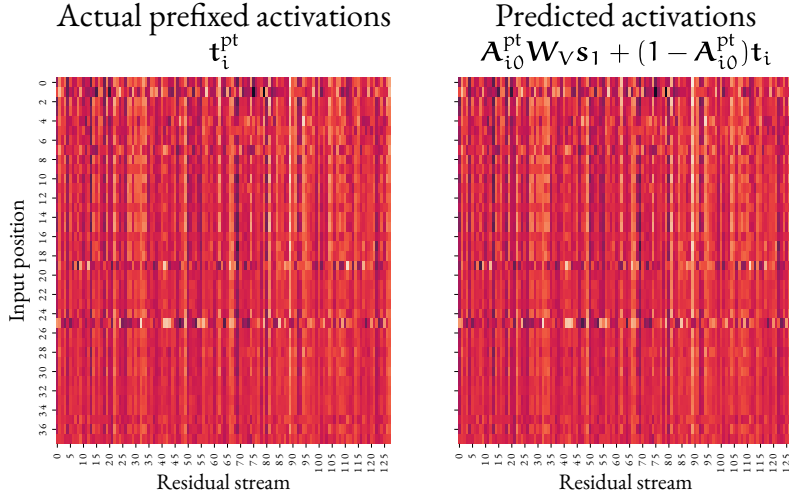


Figure 6: The activations of the twelfth head of the first layer of LLaMA (Touvron et al., 2023). The left plot shows the activations in the presence of the prefix. The right plot shows the activations $\mathbf{t}_i$ of the pretrained model, scaled by one minus the attention that the prefix would take and then biased in the direction $\mathbf{W}_V \mathbf{s}_1$. The two plots are the same, illustrating that our theory, Equation (7) in particular, also holds for real-world large transformer models. The test sequence is the same as in Figure 5.

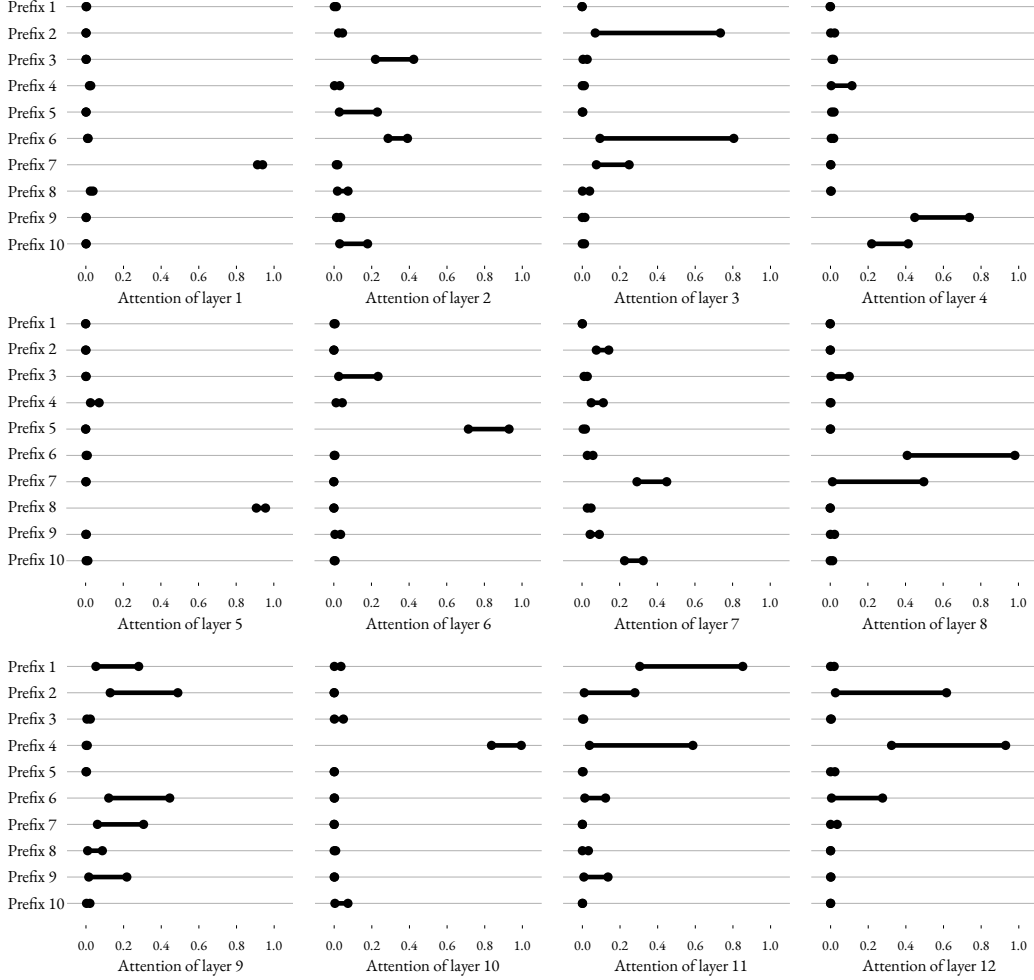


Figure 7: The range of attention (1st to 99th percentile) for a single GPT-2 (Radford et al., 2019) prefix trained on the Emotion dataset (Saravia et al., 2018). The prefix is of size 10 ($n_S = 10$). This is the attention of the last user input token (n_X) because this is the position at which the class prediction is done. For illustration purposes, we have normalized the attention so that the attention over the 10 prefix positions sums to 1. The range of attention over the 10 positions for each layer are shown.

B ATTENTION DISTRIBUTION OVER THE PREFIX

As discussed in Section 4, longer prefixes define a subspace from which the bias for the attention block is selected. For a prefix of size n_S , that means that this subspace is n_S -dimensional. Each of prefix position j defines a basis vector $\mathbf{W}_V \mathbf{s}_j$ for this subspace, while the attention A_{i,S_j}^{pt} on this position determines how much of this basis component contributes to the bias.

In order to span the whole subspace and make full use of the capacity of the prefix, A_{i,S_j}^{pt} should vary between 0 and 1 for different inputs. However, we observe that this does not happen in practice. Figure 7 shows the ranges of attention the different prefix positions take for the GPT-2 model (Radford et al., 2019). For layer 1, for example, the attention each prefix positions gets is almost constant hence, the effective subspace is collapsed and there is a single bias vector that’s applied to the attention layer output, regardless of the user input X .

Some other layers show slightly higher variation. For example, layer 3 has three prefix positions with large variations. Therefore, the effective bias subspace is 3-dimensional and the user input X governs which bias vector from this subspace will be selected.

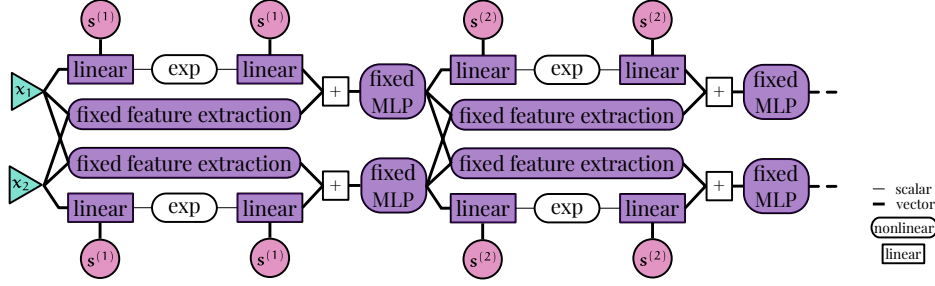


Figure 8: Prefix-tuning as a neural network architecture. While linearities and non-linearities are present, the only learnable parameters $s^{(1)}, s^{(2)}, \dots$ have limited interaction with the inputs x_1 and x_2 . The interaction of the prefix parameters with each input is only via the scalar attention, shown here with a light connection. The mixing of information between the inputs happens via residual connections with the pretrained fixed feature extraction and hence is not learnable. The MLP is also fixed and hence only acts as a multivariate activation function. This limited interaction explains why prefix-tuning struggles to learn new tasks even in deeper models.

C EXPRESSIVITY OF PREFIX-TUNING ACROSS DEEPER MODELS

In Section 6, we considered the effect of the presence of the prefix in the first attention layer on the attention of the second. However, the effects become more complex as one adds more attention layers. We also ignored the MLPs between, but in practice they can play an important role. Here, instead, we analyse prefix-tuning as learning a neural network. We argue that while the resulting architecture includes both linear operation and non-linear activations, the structure is unlikely to learn efficiently.

For simplicity, we will consider two inputs x_1 and x_2 and a single prefix s . The output of the attention head, parameterized by s is then:

$$\begin{aligned} \mathcal{A}_s(x_1, x_2) &= \langle y_1, y_2 \rangle \\ y_1 &= \frac{\exp(x_1^\top H s) W_V s + \exp(x_1^\top H x_1) W_V x_1 + \exp(x_1^\top H x_2) W_V x_2}{C_1} \\ y_2 &= \frac{\exp(x_2^\top H s) W_V s + \exp(x_2^\top H x_1) W_V x_1 + \exp(x_2^\top H x_2) W_V x_2}{C_2} \end{aligned} \quad (11)$$

where we have omitted the $T/\sqrt{k}$ factors and have folded the softmax normalization into C_1 and C_2 . The **layer inputs**, **pretrained parameters** and the **learnable parameters** are correspondingly highlighted. The attention head is clearly a non-linear operation. However, the learnable parameter s participates only in the left term. It interacts with only one of the inputs at a time and only by computing a single scalar value $x_i^\top H s$. As we discussed above, the interaction between x_1 and x_2 is non-trainable and can be thought as a hard-coded feature extraction. Each of the outputs is then passed to a pretrained MLP which can be thought of as an activation function. This would be a *multivariate* activation function, which while unusual in the contemporary practice has been studied before (Solazzi and Uncini, 2004). Figure 8 illustrates the computation graph of the resulting neural network and shows that the only learnable interaction between the inputs is indirect. Therefore, prefix-tuning can be considered as learning a neural network where the only interaction between the inputs happens via non-learnable residual connections. Nevertheless, the alternating linear and nonlinear operations are reminiscent of the standard neural network architecture and their universal approximation properties (Hassoun, 1995). That begs the question if the prefix-tuning architecture can be a universal approximator and whether it would be a parameter-efficient one.

An example of prefix-tuning failing to be a universal approximator. While we leave the formal analysis of the representational capacity of prefix-tuning as future work, we provide an example of pretrained parameters for which the architecture is *not* a universal approximator. As can be seen in Figure 8, all information must pass through the non-learnable MLPs. Thus, if the MLPs destroy all input information, there is no value for the prefixes $s^{(1)}, s^{(2)}, \dots$ that can change that fact. The MLPs